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Synthesis and effect of silver nanoparticles on the antibacterial activity of different antibiotics against *Staphylococcus aureus* and *Escherichia coli*

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Abstract Silver nanoparticles (Ag-NPs) have been known to have inhibitory and bactericidal effects. Resistance to antimicrobial agents by pathogenic bacteria has emerged in recent years and is a major health problem. The combination effects of Ag-NPs with the antibacterial activity of antibiotics have not been studied. Here, we report on the synthesis of metallic nanoparticles of silver using a reduction of aqueous Ag^+ ion with the culture supernatants of *Klebsiella pneumoniae*. Also in this article these nanoparticles are evaluated for their part in increasing the antimicrobial activities of various antibiotics against *Staphylococcus aureus* and *Escherichia coli*. The antibacterial activities of penicillin G, amoxicillin, erythromycin, clindamycin, and vancomycin were increased in the presence of Ag-NPs against both test strains. The highest enhancing effects were observed for vancomycin, amoxicillin, and penicillin G against *S. aureus.*

Key words: Antibiotics resistance; *Klebsiella pneumoniae*; Silver nanoparticles; Synthesis

Human beings are often infected by microorganisms such as bacteria, molds, yeasts, and viruses in the living environment. Research in antibacterial material containing various natural and inorganic substances [1,2] has been intensive. Metal nanoparticles (Me-NPs), which have a high specific surface area and a high fraction of surface atoms, have been studied extensively because of their unique physicochemical characteristics including catalytic activity, optical properties, electronic properties, antimicrobial activity, and magnetic properties [3-5]. Among Me-NPs, silver nanoparticles (Ag-NPs) have been known to have inhibitory and bactericidal effects [2]. It can be expected that the high specific surface area and high fraction of surface atoms of Ag-NPs will lead to high antimicrobial activity as compared with bulk silver metal [2]. The combined effects of Ag-NPs with the antibacterial activity of antibiotics have not been studied. The ability of pathogenic bacteria to resist antimicrobial agents has emerged in recent years and is a major health problem [6,7]. In this study, Ag-NPs were evaluated for use in increasing the antimicrobial activities of different antibiotics against *S. aureus* and *E. coli*.

Materials and methods

Synthesis of Ag-NPs

Colloidal Ag-NPs solution was prepared following the method already described [8,9]. Müller-Hinton medium was prepared, sterilized, and inoculated with a fresh growth of test strain. The cultured flasks were incubated at 37°C for 24 hours. After the incubation time the culture was centrifuged at 12,000 rpm and the supernatant was used for

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Fig 1. Conical flasks with silver nitrate (1 mM) before (left) and after (right) exposure to the culture supernatant of *Klebsiella pneumoniae* (Enterobacteriaceae).

the synthesis of Ag-NPs. *K. pneumoniae* culture supernatant was separately added to the reaction vessels containing silver nitrate at a concentration of 10^{-3} M (1% v/v). The reaction between this supernatant and Ag^+ ions was carried out in bright conditions for 5 minutes. The bioreduction of the Ag^+ ions in the solution was monitored by sampling the aqueous component (2 mL) and measuring the ultraviolet-visible (UV-vis) spectrum of the solution. UV-vis spectra of these samples were monitored on a Cecil model 9200 UV-vis spectrophotometer, operated at a resolution of 1 nm. Furthermore, the Ag-NPs were characterized by transmission electron microscopy (model EM 208 Philips, Eindhoven, The Netherlands) and energy-dispersive spectroscopy.

Disk diffusion assay to evaluate combined effects

A disk diffusion method was used to assay the various antibiotics for bactericidal activity against test strains on Müller-Hinton agar plates. The standard antibiotics disks were purchased from Mast Co. (Liverpool, UK). To determine the combined effects, each standard paper disk was further impregnated with 10 μL of the freshly prepared Ag-NPs at a final content of 10 μg/disk. A single colony of each test strain was grown overnight in Müller-Hinton liquid medium on a rotary shaker (200 rpm) at 35°C. The inocula were prepared by diluting the overnight cultures with 0.9% NaCl to a 0.5 McFarland standard and were applied to the plates along with the standard and prepared disks containing differing amounts of Ag-NPs. Clinical isolates of *S. aureus* and *E. coli* from our culture collection were used as test strains. Similar experiments were carried out with Ag-NPs alone. After incubation at 35°C for 18 hours the zones of inhibition were measured. The assays were performed in triplicate.

Results and discussion

Ag-NPs were synthesized from Ag^+ ions by treating the supernatant of *K. pneumonia*. The appearance of a

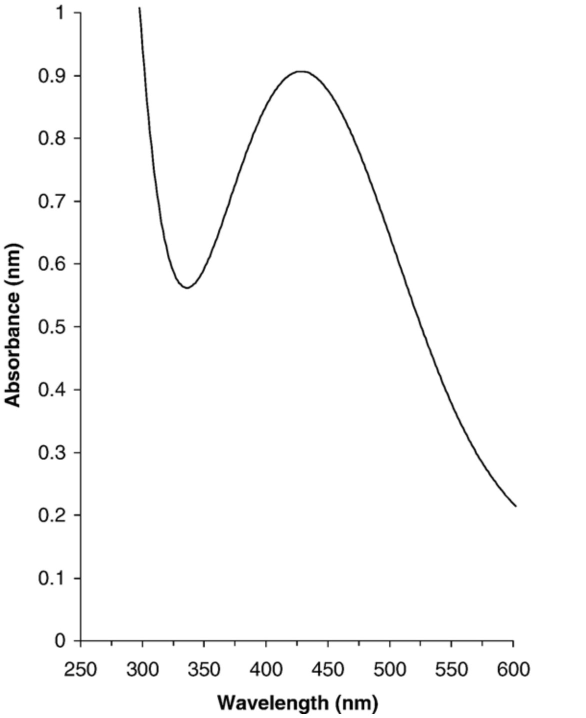

Fig 2. Ultraviolet-visible spectra recorded after the addition of culture supernatant of *Klebsiella pneumoniae* (1 mL) to 1 mM silver nitrate solution (100 mL). The curves are recorded after a period of 5 minutes. The test strain was cultivated in Muller-Hinton broth and was incubated at 35°C for 24 hours. After the incubation period, the culture was centrifuged at 12,000 rpm, and its supernatants were used to reduce the silver nitrate solution.

yellowish brown color in the reaction vessels suggested the formation of Ag-NPs [10]. The supernatant of the *K. pneumonia* culture changed the solution to a brownish color upon the completion of the 5-minute reaction with Ag^+ (Figure 1). The Ag-NPs were characterized by UV-vis spectroscopy. The technique outlined above has proved to be very useful for the analysis of nanoparticles [11-13]. As illustrated in Uv-vis spectra, a strong surface plasmon resonance was centered at approximately 430 nm (Figure 2). Observation of this strong but broad surface plasmon peak has been well documented for various Me-NPs, with sizes ranging all the way from 2 to 100 nm [11-13]. Figure 3 shows a representative SEM image recorded from the drop-coated film of the Ag-NPs that were synthesized by treating the silver nitrate solution with culture supernatant of *K. pneumonia*. The particle size histograms of silver particles (right-hand illustration in Figure 3) show that the particles range in size from 5 to 32 nm and possess an average size of 22.5 nm.

In our analysis by energy-dispersive spectroscopy of the Ag-NPs we confirmed the presence of elemental silver signal (Figure 4). The silver nanocrystallites display an optical absorption band peak at approximately 3 KeV, which is typical of the absorption of metallic silver nanocrystallites due to surface plasmon resonance [14].

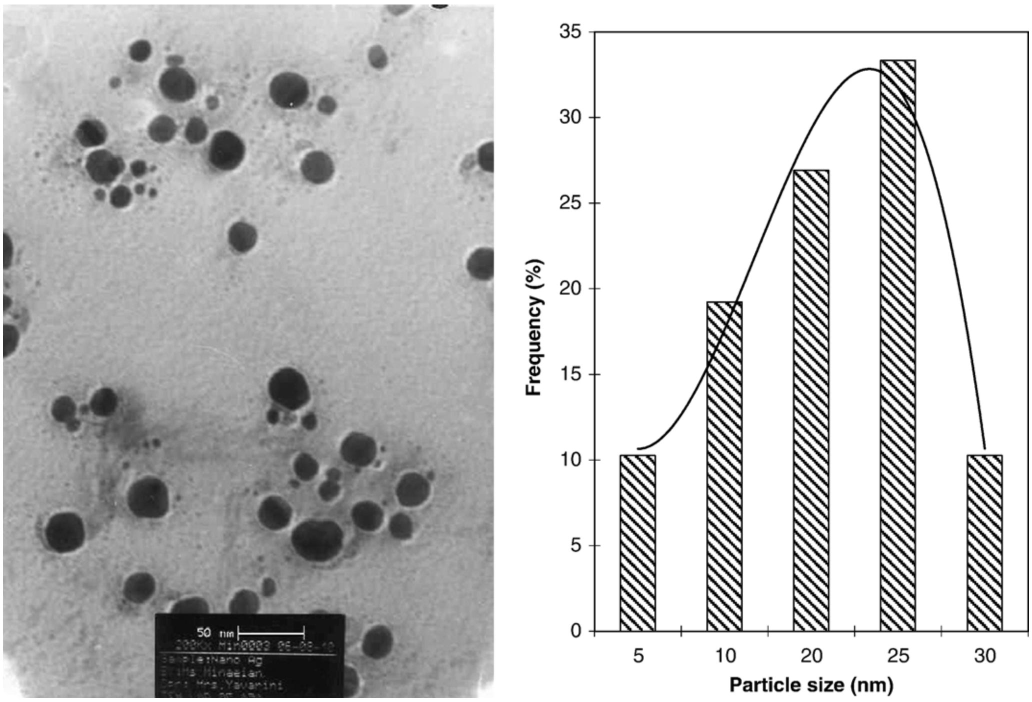

Fig 3. Transmission electron micrographs of silver nanoparticles formed by reducing Ag^+ ions using the culture supernatants of enterobacteria. Scale bars correspond to 50 nm. Particle size histogram of the silver particles is shown in the righthand picture.

NfsA, the major oxygen-insensitive nitroreductase of Enterobacteriacae, is a flavoprotein that is able to reduce nitro groups on many different nitroaromatic compounds under aerobic conditions [15]. Also this enzyme is responsible for reduction of chromate to less soluble and less toxic Cr_{III} [16]. The nitroreductase enzymes may be involved in Ag^+ reduction process.

The combination effect of these nanoparticles with different antibiotics was investigated against *S. aureus* and *E. coli* using the disk diffusion method. The diameter of inhibition zones (in millimeters) around the different antibiotic disks with or without Ag-NPs against test strains are shown in Table 1.

The antibacterial activities of penicillin G, amoxicillin, erythromycin, clindamycin, and vancomycin increased in the presence of Ag-NPs against both test strains. No enhancing effect on the antibacterial activities of other antibiotics was observed against *S. aureus* and *E. coli* at the concentrations tested. The highest fold increases in area were observed for vancomycin, amoxicillin, and penicillin G against *S. aureus*. The effects of Ag-NPs on the antibacterial activity of the aforementioned antibiotics for *E. coli* were lower than *S. aureus*. In contrast, the most synergistic activity was observed with erythromycin against S. *aureus*.

Chemical synthesis methods may lead to the presence of some toxic chemical species adsorbed on the surface that may have adverse effects in its application [10]. On the other hand, the possible chemical residues on nanoparticles may interact with biological systems as well as the antibacterial effects of antibiotics. However, synthesis of nanoparticles using microorganisms can potentially eliminate this problem by making the nanoparticles more biocompatible, so we used this strategy for the present investigation. At this time the reason for these developments and the reason for these differences are not known and merits investigation. This is the first report concerning the enhancement of the activity of some antibiotics by Ag-NPs synthesized by supernatants of *K. pneumonia*.

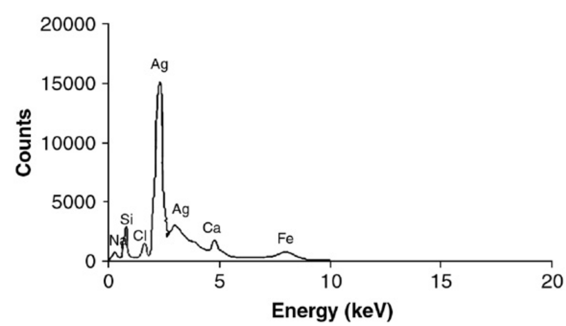

Fig 4. Energy-dispersive spectroscopy spectrum of silver nanoparticles. Various x-ray emission peaks are labeled.

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Table 1
Zone of inhibition (mm) of different antibiotics against test strains (in absence and in presence of Ag-NPs at content of 10 μg/disk)

Antibiotics (μg/disk)	*Staphylococcus aureus*		Increase in fold area†	*Escherichia coli*		Increase in fold area†
	Antibiotic only (A)	Antibiotic plus Ag-NPs* (B)		Antibiotic only (C)	Antibiotic plus Ag-NPs* (D)	
Penicillin G 10	–‡	12	1.938	8	12	1.25
Amoxicillin 10	7.5	14	2.484	10	12	0.44
Carbenicillin 100	28	28	0	26	26	0
Cephalexin 30	16	16	0	15	15	0
Cefixime 5	29	29	0	28	28	0
Erythromycin 5	10	14	0.96	8	12	1.25
Gentamicin 10	25	25	0	24	24	0
Amikacin 30	23	23	0	27	27	0
Tetracycline 30	25	25	0	25	25	0
Co-trimoxazole 25	36	36	0	37	37	0
Clindamycin 2	–	9	0.653	–	9	0.653
Nitrofurantoin 300	25	25	0	23	23	0
Nalidixic acid 30	25	25	0	25	25	0
Vancomycin 30	–	13	2.45	–	10	1.04

* The mean of inhibition zone diameter around the disk containing Ag-NPs alone (10 μg) was 9 mm. All experiments were done in triplicate, and standard deviations were negligible.

† Mean surface area of the inhibition zone (mm^2) was calculated for each tested antibiotic from the mean diameter. Fold increases for different antibiotics against *S. aureus* were calculated as $(b^2 - a^2)/a^2$, where a and b are the inhibition zones for A and B, respectively. In the same way, $(d^2 - c^2)/c^2$ was used for antibiotics against *E. coli*.

‡ In the absence of bacterial growth inhibition zones, the disks' diameters (7 mm) were used to calculate the fold increase in columns 3 and 6.

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